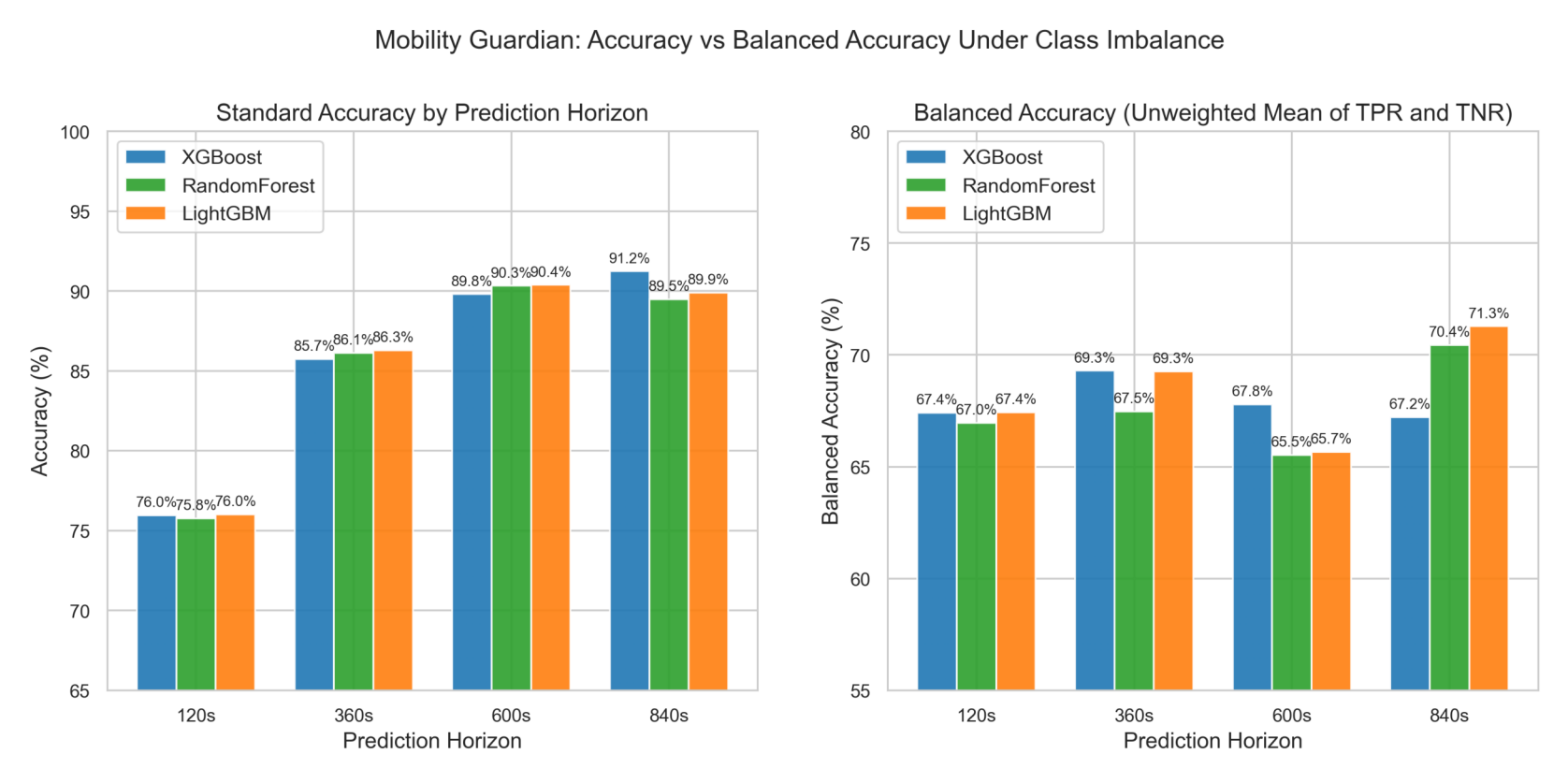
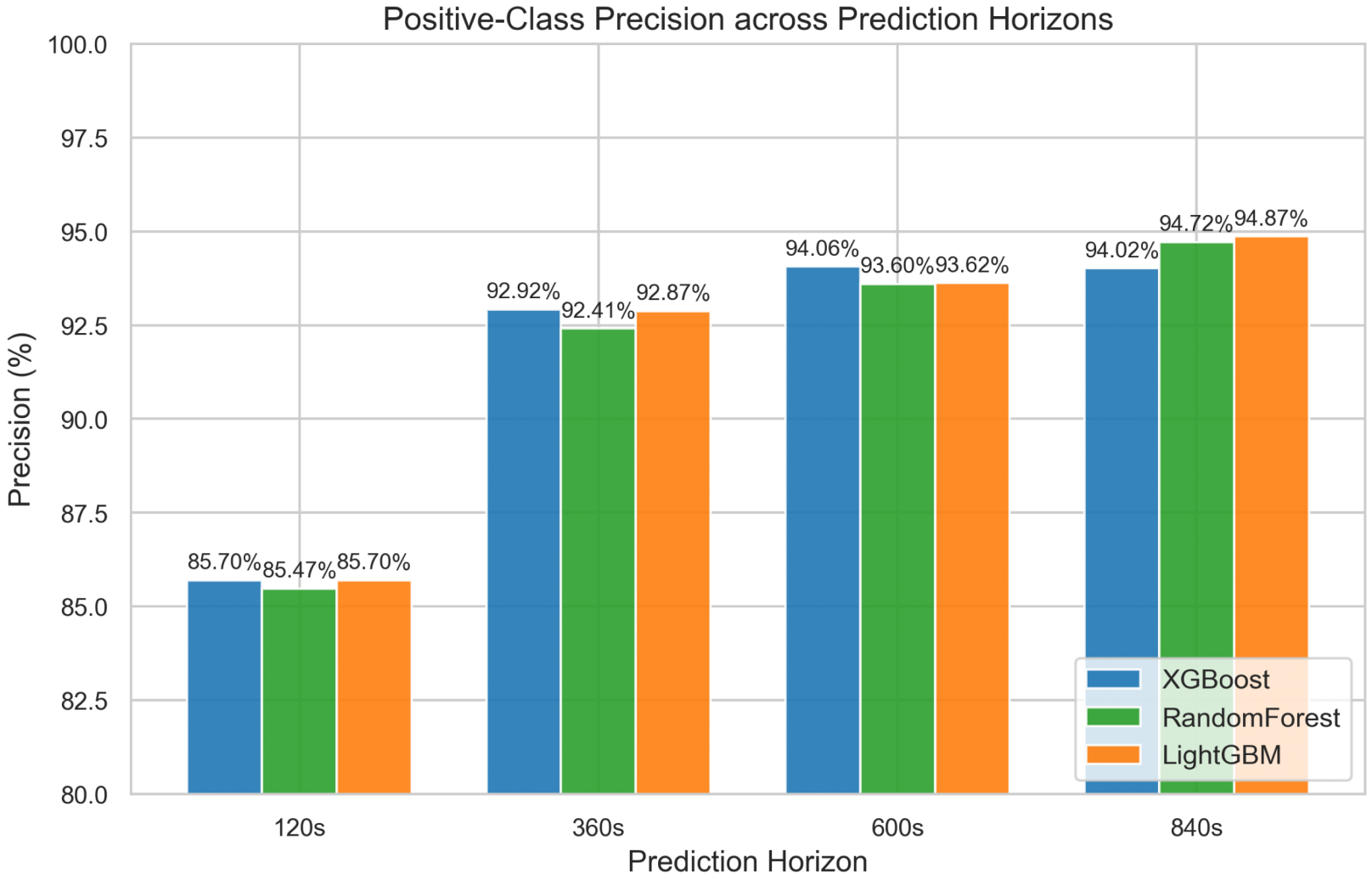
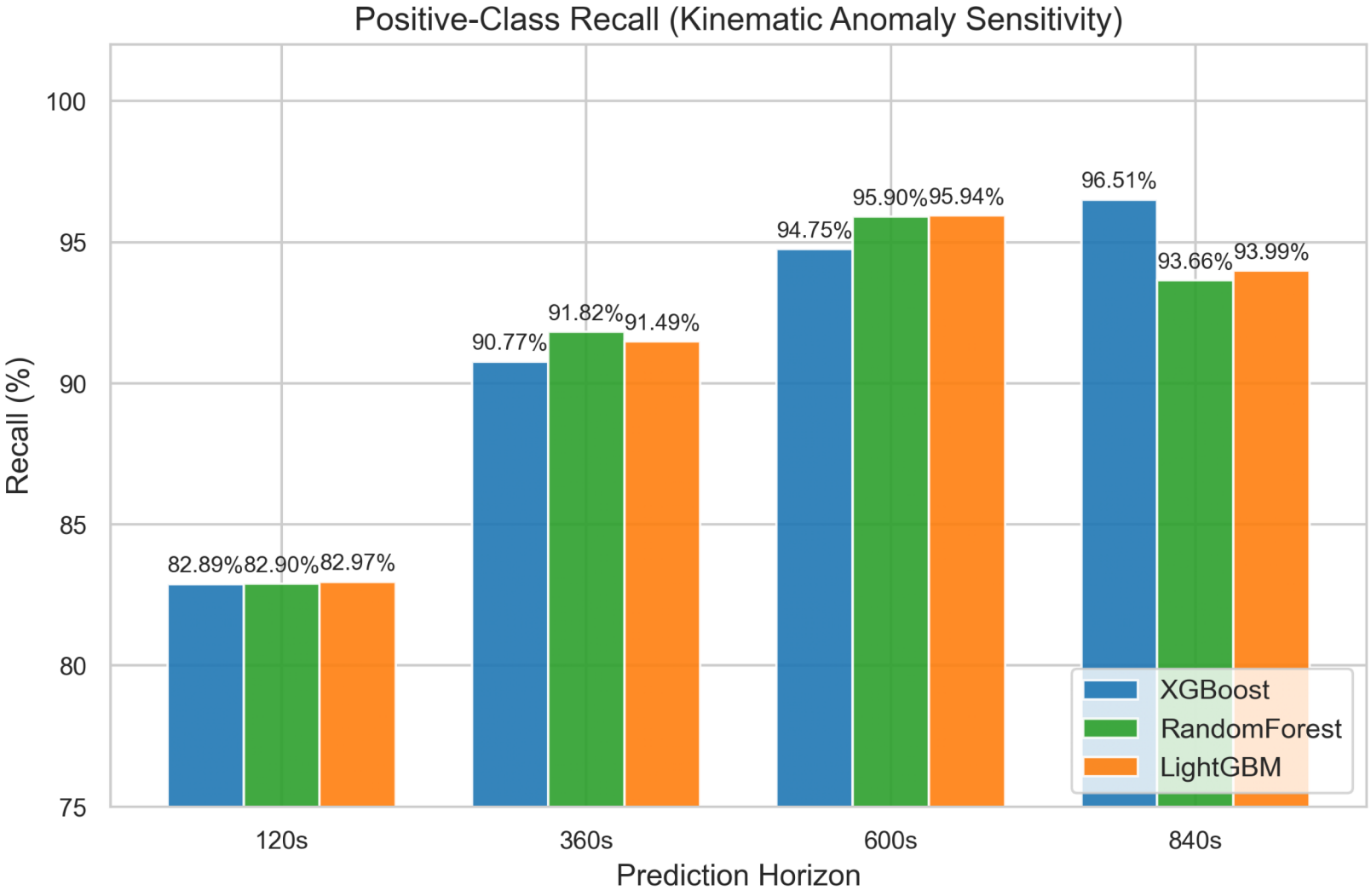
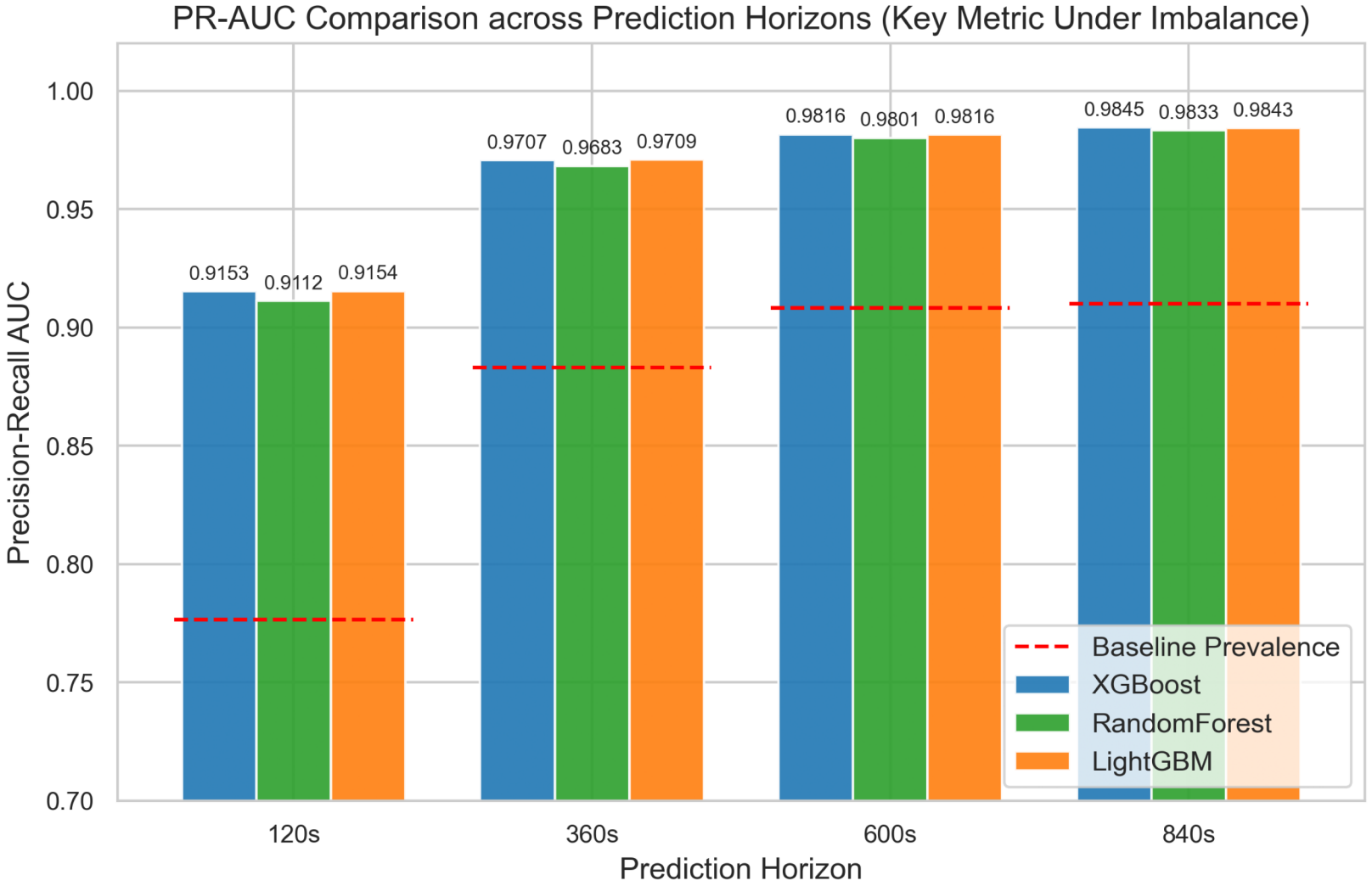
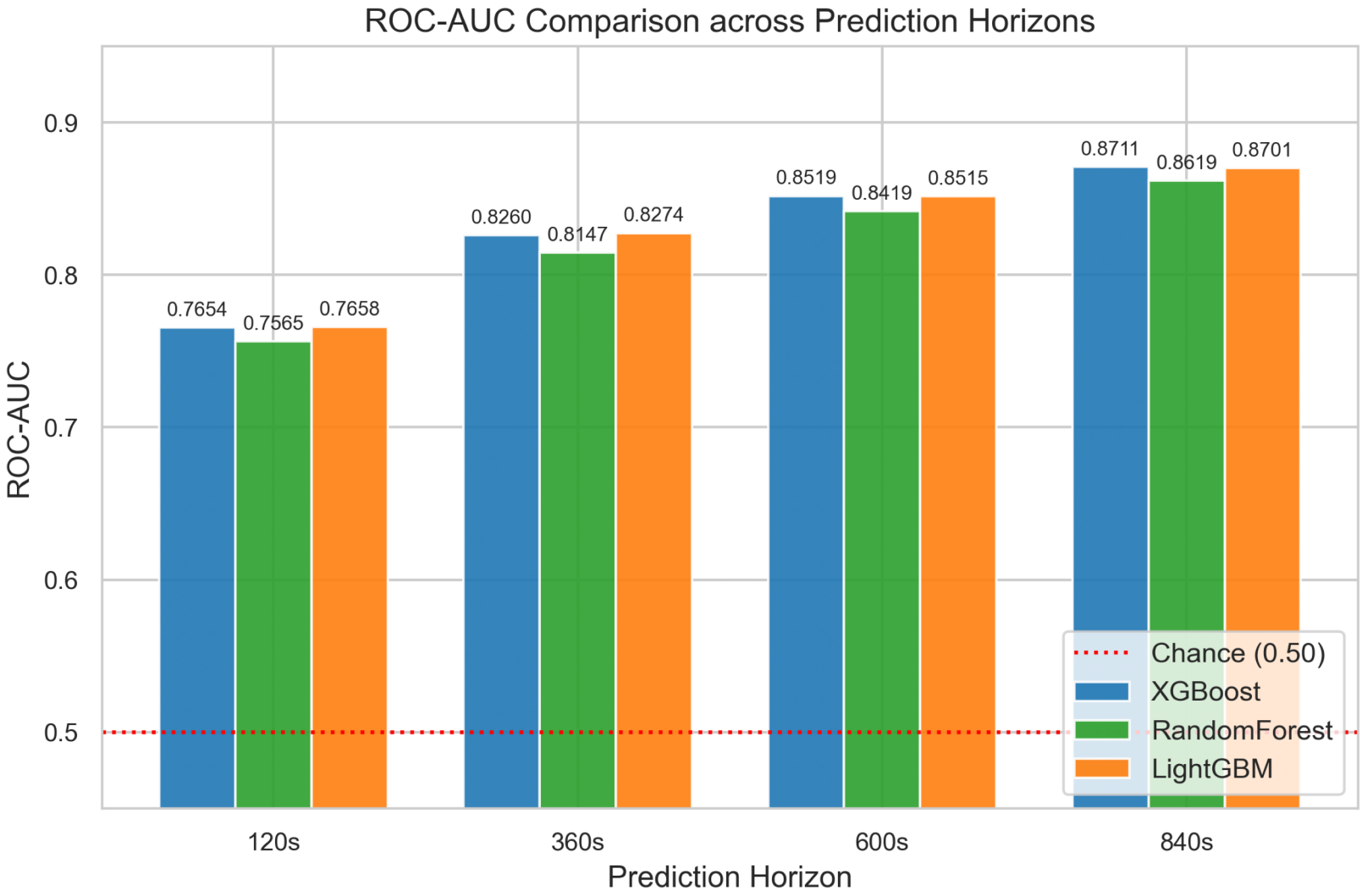

Mobility Guardian: Accuracy vs Balanced Accuracy Under Class Imbalance
Standard Accuracy by Prediction Horizon
XGBoost
RandomForest
LightGBM
Accuracy (%)
100
95
90
85
80
75
70
65
76.0%
75.8%
76.0%
85.7%
86.1%
86.3%
89.8%
90.3%
90.4%
91.2%
89.5%
89.9%
120s
360s
600s
840s
Prediction Horizon
Balanced Accuracy (Unweighted Mean of TPR and TNR)
XGBoost
RandomForest
LightGBM
Balanced Accuracy (%)
80
75
70
65
60
55
67.4%
67.0%
67.4%
69.3%
67.5%
69.3%
67.8%
65.5%
65.7%
67.2%
70.4%
71.3%
120s
360s
600s
840s
Prediction Horizon

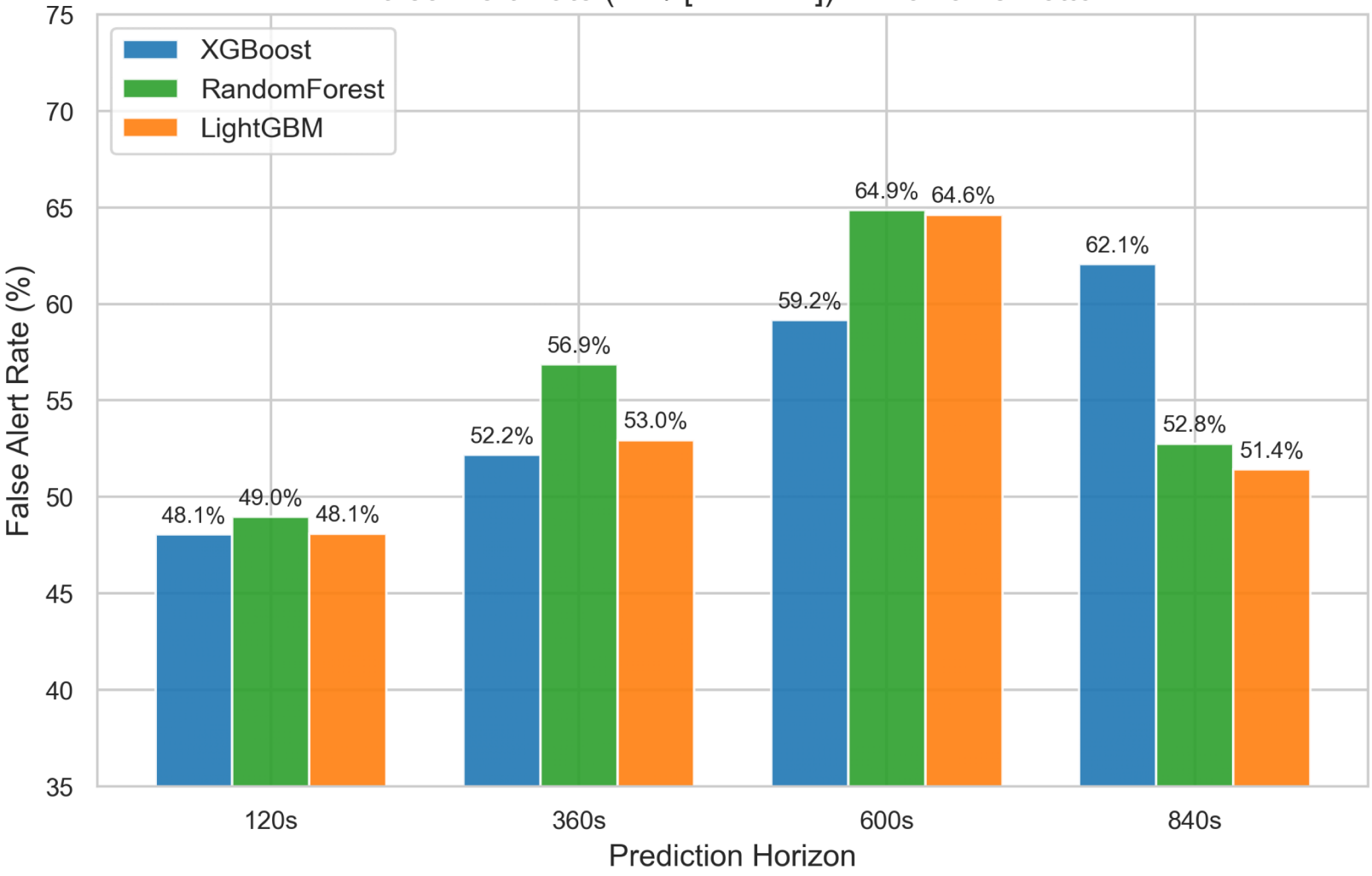

Positive-Class Precision across Prediction Horizons
Precision (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
85.70%
85.47%
85.70%
92.92%
92.41%
92.87%
94.06%
93.60%
93.62%
94.02%
94.72%
94.87%
120s
360s
600s
840s
Prediction Horizon
XGBoost
RandomForest
LightGBM

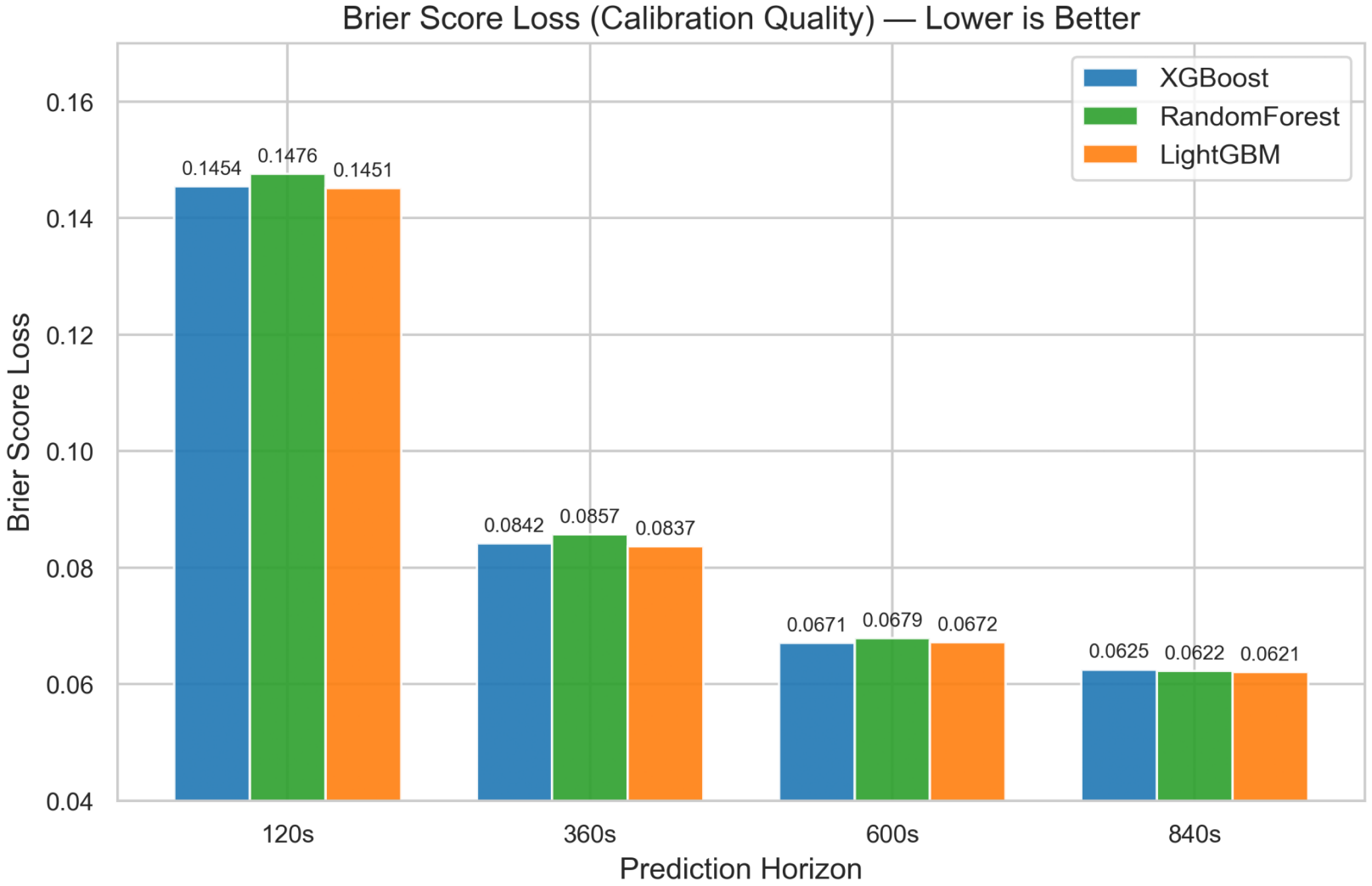

Positive-Class Recall (Kinematic Anomaly Sensitivity)
Recall (%)
100
95
90
85
80
75
82.89%
82.90%
82.97%
90.77%
91.82%
91.49%
94.75%
95.90%
95.94%
96.51%
93.66%
93.99%
XGBoost
RandomForest
LightGBM
120s
360s
600s
840s
Prediction Horizon

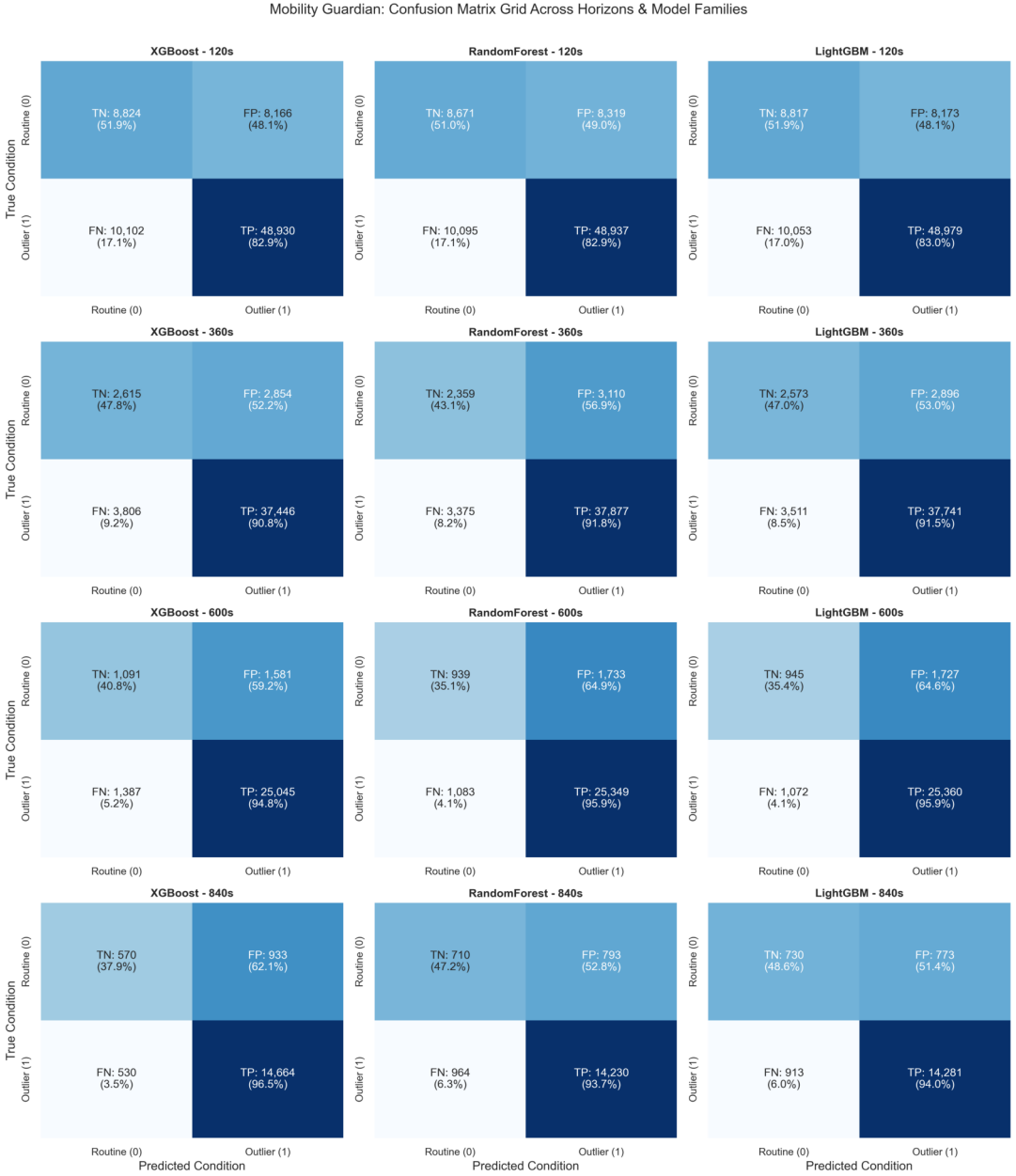

Mobility Guardian: F1 Score Comparison
Positive-Class F1 Score (Risk/Outlier Class)
F1 Score
Prediction Horizon
120s
360s
600s
840s
0.843
0.842
0.843
0.918
0.921
0.922
0.944
0.947
0.948
0.952
0.942
0.944
XGBoost
RandomForest
LightGBM
Macro-F1 Score (Balanced Across Both Classes)
Macro-F1 Score
0.667
0.663
0.667
0.679
0.671
0.684
0.684
0.674
0.675
0.695
0.694
0.704

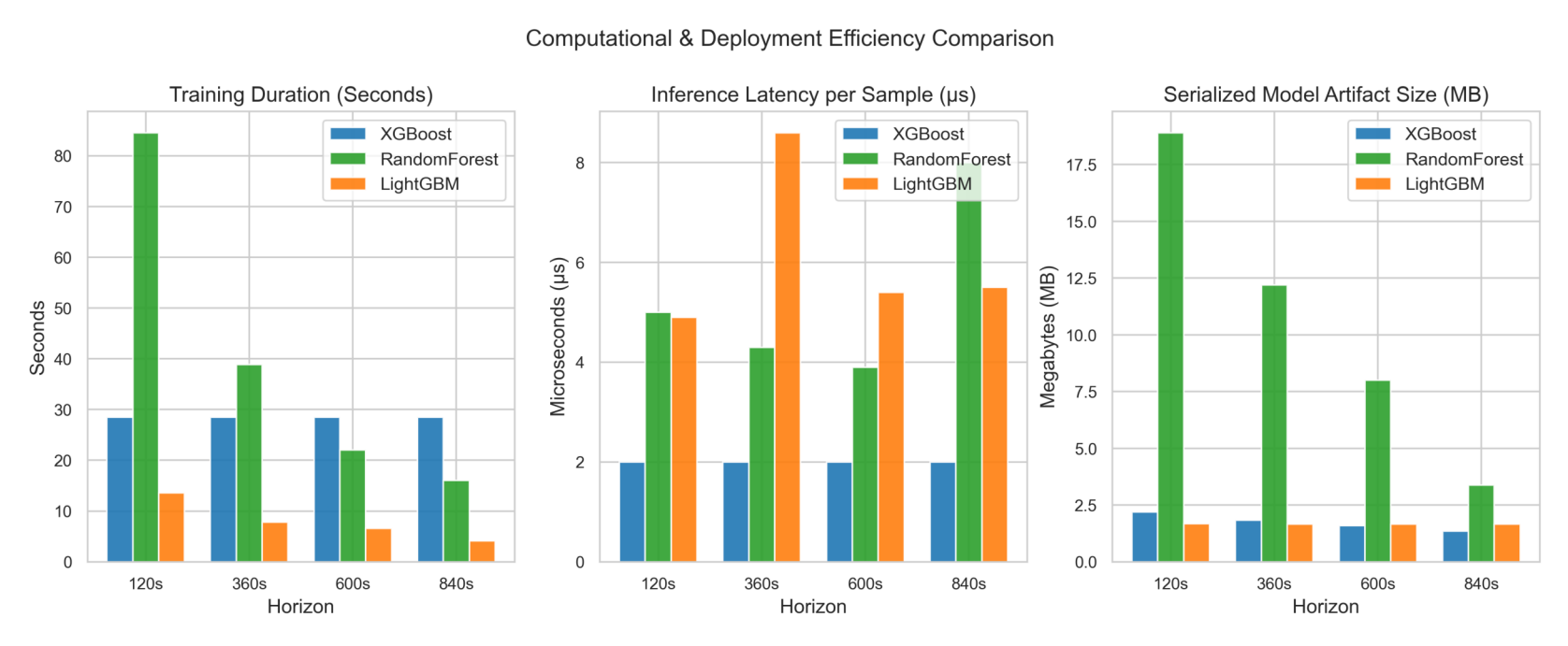

PR-AUC Comparison across Prediction Horizons (Key Metric Under Imbalance)
Precision-Recall AUC
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.9153
0.9112
0.9154
0.9707
0.9683
0.9709
0.9816
0.9801
0.9816
0.9845
0.9833
0.9843
120s
360s
600s
840s
Prediction Horizon
Baseline Prevalence
XGBoost
RandomForest
LightGBM

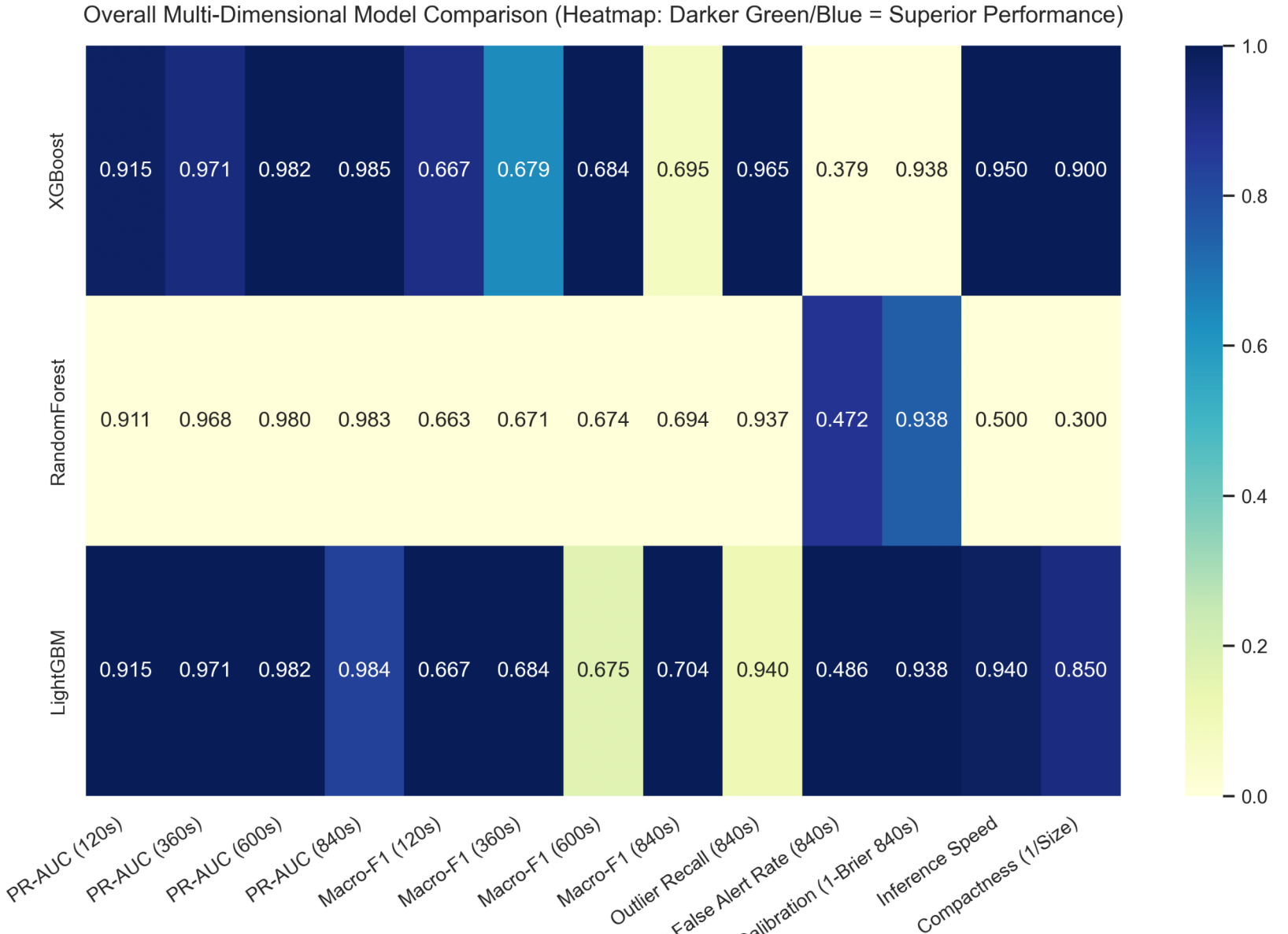

ROC-AUC Comparison across Prediction Horizons
ROC-AUC
0.9
0.8
0.7
0.6
0.5
0.7654
0.7565
0.7658
0.8260
0.8147
0.8274
0.8519
0.8419
0.8515
0.8711
0.8619
0.8701
Chance (0.50)
XGBoost
RandomForest
LightGBM
120s
360s
600s
840s
Prediction Horizon


False Alert Rate (FP / [FP + TN]) — Lower is Better
XGBoost
RandomForest
LightGBM
False Alert Rate (%)
75
70
65
60
55
50
45
40
35
48.1%
49.0%
48.1%
52.2%
56.9%
53.0%
59.2%
64.9%
64.6%
62.1%
52.8%
51.4%
120s
360s
600s
840s
Prediction Horizon

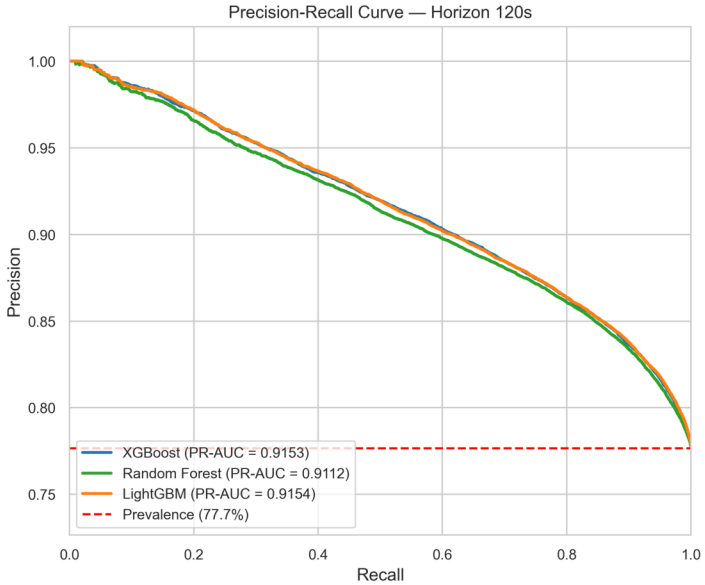

Brier Score Loss (Calibration Quality) — Lower is Better
XGBoost
RandomForest
LightGBM
0.16
0.14
0.12
0.10
0.08
0.06
0.04
Brier Score Loss
0.1454
0.1476
0.1451
0.0842
0.0857
0.0837
0.0671
0.0679
0.0672
0.0625
0.0622
0.0621
120s
360s
600s
840s
Prediction Horizon

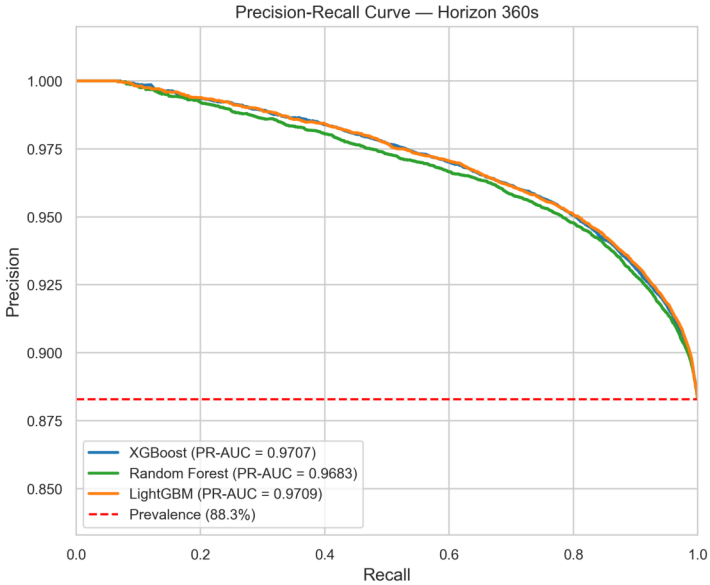

Mobility Guardian: Confusion Matrix Grid Across Horizons & Model Families
XGBoost - 120s
TN: 8,824 (51.9%)
FP: 8,166 (48.1%)
FN: 10,102 (17.1%)
TP: 48,930 (82.9%)
RandomForest - 120s
TN: 8,671 (51.0%)
FP: 8,319 (49.0%)
FN: 10,095 (17.1%)
TP: 48,937 (82.9%)
LightGBM - 120s
TN: 8,817 (51.9%)
FP: 8,173 (48.1%)
FN: 10,053 (17.0%)
TP: 48,979 (83.0%)
XGBoost - 360s
TN: 2,615 (47.8%)
FP: 2,854 (52.2%)
FN: 3,806 (9.2%)
TP: 37,446 (90.8%)
RandomForest - 360s
TN: 2,359 (43.1%)
FP: 3,110 (56.9%)
FN: 3,375 (8.2%)
TP: 37,877 (91.8%)
LightGBM - 360s
TN: 2,573 (47.0%)
FP: 2,896 (53.0%)
FN: 3,511 (8.5%)
TP: 37,741 (91.5%)
XGBoost - 600s
TN: 1,091 (40.8%)
FP: 1,581 (59.2%)
FN: 1,387 (5.2%)
TP: 25,045 (94.8%)
RandomForest - 600s
TN: 939 (35.1%)
FP: 1,733 (64.9%)
FN: 1,083 (4.1%)
TP: 25,349 (95.9%)
LightGBM - 600s
TN: 945 (35.4%)
FP: 1,727 (64.6%)
FN: 1,072 (4.1%)
TP: 25,360 (95.9%)
XGBoost - 840s
TN: 570 (37.9%)
FP: 933 (62.1%)
FN: 530 (3.5%)
TP: 14,664 (96.5%)
RandomForest - 840s
TN: 710 (47.2%)
FP: 793 (52.8%)
FN: 964 (6.3%)
TP: 14,230 (93.7%)
LightGBM - 840s
TN: 730 (48.6%)
FP: 773 (51.4%)
FN: 913 (6.0%)
TP: 14,281 (94.0%)
True Condition
Routine (0)
Outlier (1)
Predicted Condition

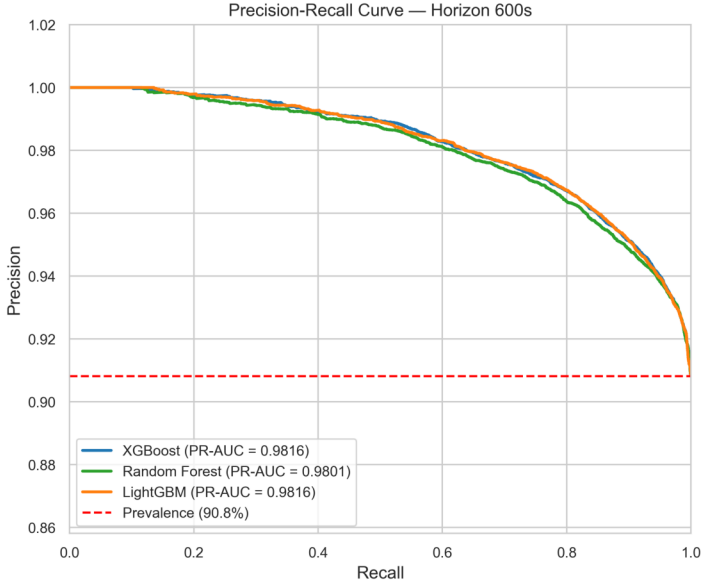

Computational & Deployment Efficiency Comparison
Training Duration (Seconds)
Seconds
Horizon
120s
360s
600s
840s
XGBoost
RandomForest
LightGBM
Inference Latency per Sample (µs)
Microseconds (µs)
Horizon
Serialized Model Artifact Size (MB)
Megabytes (MB)
Horizon

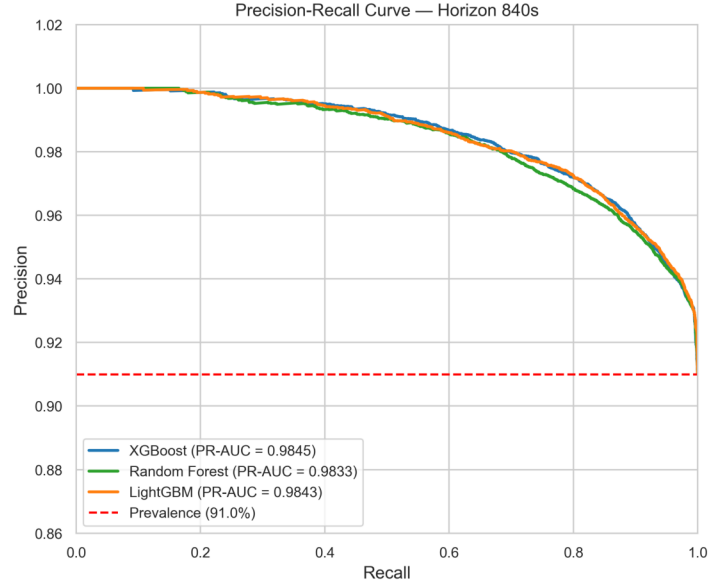

Overall Multi-Dimensional Model Comparison (Heatmap: Darker Green/Blue = Superior Performance)
XGBoost
RandomForest
LightGBM
0.915 0.971 0.982 0.985 0.667 0.679 0.684 0.695 0.965 0.379 0.938 0.950 0.900
0.911 0.968 0.980 0.983 0.663 0.671 0.674 0.694 0.937 0.472 0.938 0.500 0.300
0.915 0.971 0.982 0.984 0.667 0.684 0.675 0.704 0.940 0.486 0.938 0.940 0.850
PR-AUC (120s)
PR-AUC (360s)
PR-AUC (600s)
PR-AUC (840s)
Macro-F1 (120s)
Macro-F1 (360s)
Macro-F1 (600s)
Macro-F1 (840s)
Outlier Recall (840s)
Low False Alert Rate (840s)
Calibration (1-Brier 840s)
Inference Speed
Compactness (1/Size)
1.0
0.8
0.6
0.4
0.2
0.0

Actual Precision-Recall Curves from Held-Out Test Cohort Predictions









Actual ROC Curves from Held-Out Test Cohort Predictions

